Mekelle University

College of dry land agriculture and natural resource

Department of Agricultural and resource Economics

Program Natural Resource Economics and Management

individual Assignment (15%)

1. **Why econometrics is important for Natural Resource Economics and Management (NREM) graduates?**
2. **Explain the meaning of the following properties in any regression analysis**
   i) Consistence
   ii) Efficient
   iii) Unbiased
3. ***From the following table answer the questions accordingly!***
   A. Calculate the values of **A, B, C**and **D. (0.5 point each)**
   B. Write the estimated equation and interpret the coefficient (2 pts.)

Reg rawplantin training				
Source	SS	Df	MS	Number of obs = 270 F (1, 268) =**A** Prob> F = 0.0000 R-squared = **B** Adj R-squared = **C** Root MSE =**D**
Model	23.1770115	1	23.1770115	
Residual	43.4896552	268	0.162274833	
Total	66.6666667	269	0.247831475	
Raw planting	Coef.	Std. Err.	T	P>|t|
Training	0.5875862	0.0491664	11.95	0.000
_cons	0.1724138	0.0334535	5.15	0.000

4. A**, what is heteroscedasticity?**
   B, what is the consequence of heteroscedasticity
   C, how do you know the existence of heteroscedasticity
   D, how it can be heteroscedasticity corrected
5. A. **What multi-collinearity problem/ autocorrelation?**
   B, what are the causes of autocorrelation?
   C, what are the consequences autocorrelation?
   D, how can autocorrelation be identified/possible detection tests and remedies?
6. What is the difference b/n time series and panel data set? Clearly differentiate the two models

7. What is the difference b/n logit and probit model regression? Clearly differentiate the two models
8. **From the following table answer the respective questions please**

reg income educationlevel experience age married divorced widowed sexmale

Source	SS	df	MS	
				Number of observations= 20
Model	48292843.5	7	6898977.65	F (3,16) = 16.95
Residual	4883994.46	12	406999.538	Prob> F = 0.0000
Total	53176838	19	2798780.95	R-squared = 0.9082
				*Adj R-squared = 0.9201
				Root MSE = 637.97

Income	Coef.	Std. Err.	T	P>|t|	[95% Conf. Interval]	
Education level	626.6945	77.2455	8.11	0.000	458.3909	794.998
Experience	438.1471	51.7678	8.46	0.000	26.85903	849.4351
Age	25.22847	5.7192	4.41	0.000	-44.24891	65.68732
Married	564.2118	158.933	3.55	0.0024	-1664.366	535.9426
Divorced	515.2433	65.3862	7.88	0.000	-2032.968	866.5365
Widowed	504.6370	78.2383	6.45	0.00	-1519.38	1000.366
Sex **male**	359.1917	39.2850	9.25	0.000	-821.8965	743.3265
_cons	5609.836	1204.252	4.66	0.001	-8233.677	-2985.995

The table above contains the regression output of both quantitative and qualitative variables. You collected data that determines an individual's income. Education level, age working experience, marital status and sex of the respondent were considered as explanatory variable. Remember the baseline (base categories) for marital status is single and the baseline (base categories) for sex of respondent is female. Therefore; based on the information given above please address questions below!

A. Does a model jointly significant? How
B. Clearly interpret each significant variable
C. What kind of drawback is detected form this model?